

Knowledge Graph-Enhanced Retrieval-Augmented Generation System for Academic Question Answering in a Data Science Curriculum

Course: Data Semantics Project

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Project Overview

- About:

This project presents a Knowledge Graph-Enhanced Retrieval-Augmented Generation (KG-RAG) system for academic question answering over a Data Science curriculum at the University of Milano-Bicocca.

- Data Source:

It uses course PDFs from Machine Learning, Data Management, and Data Semantics as the main knowledge source.

- Mechanism:

The system combines vector-based retrieval with a lightweight knowledge graph to improve answer quality, especially for questions involving relationships between topics and courses.

- Evaluation Goal:

It is evaluated against a baseline RAG system using a manually curated set of academic questions. The goal is to improve retrieval relevance and curriculum-aware reasoning while keeping the system simple and focused.

- Inspiration:

The system is inspired by GraphRAG and retrieval-augmented generation research, focusing on improving structured retrieval over academic content.

Project Motivation

- The main challenge is to answer curriculum-based questions from academic PDFs in a way that is accurate, grounded, and explainable.
- While standard RAG systems can retrieve semantically similar text chunks, they often fail to capture relationships between topics, such as which course covers a concept or how different concepts are connected.
- A knowledge graph helps address this limitation by introducing structured relationships that enrich retrieval with curriculum-level context.
- This project investigates whether a lightweight knowledge graph can improve retrieval quality and answer performance compared to baseline RAG, and which types of questions benefit most from graph-based assistance.
- The scope is intentionally focused on evaluating the practical value of graph structure in academic question answering, rather than building a full semantic knowledge graph system.

Research Problem

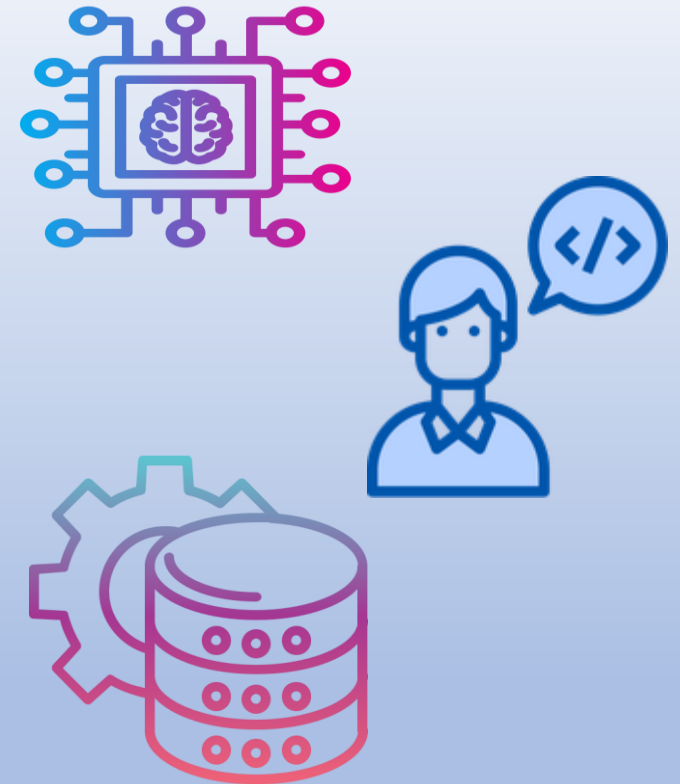
The project addresses the following questions:

- Can a lightweight knowledge graph improve retrieval for curriculum questions?
- Does graph enrichment improve answer quality compared to baseline RAG?
- Which types of questions benefit most from graph-based assistance?



Data Source

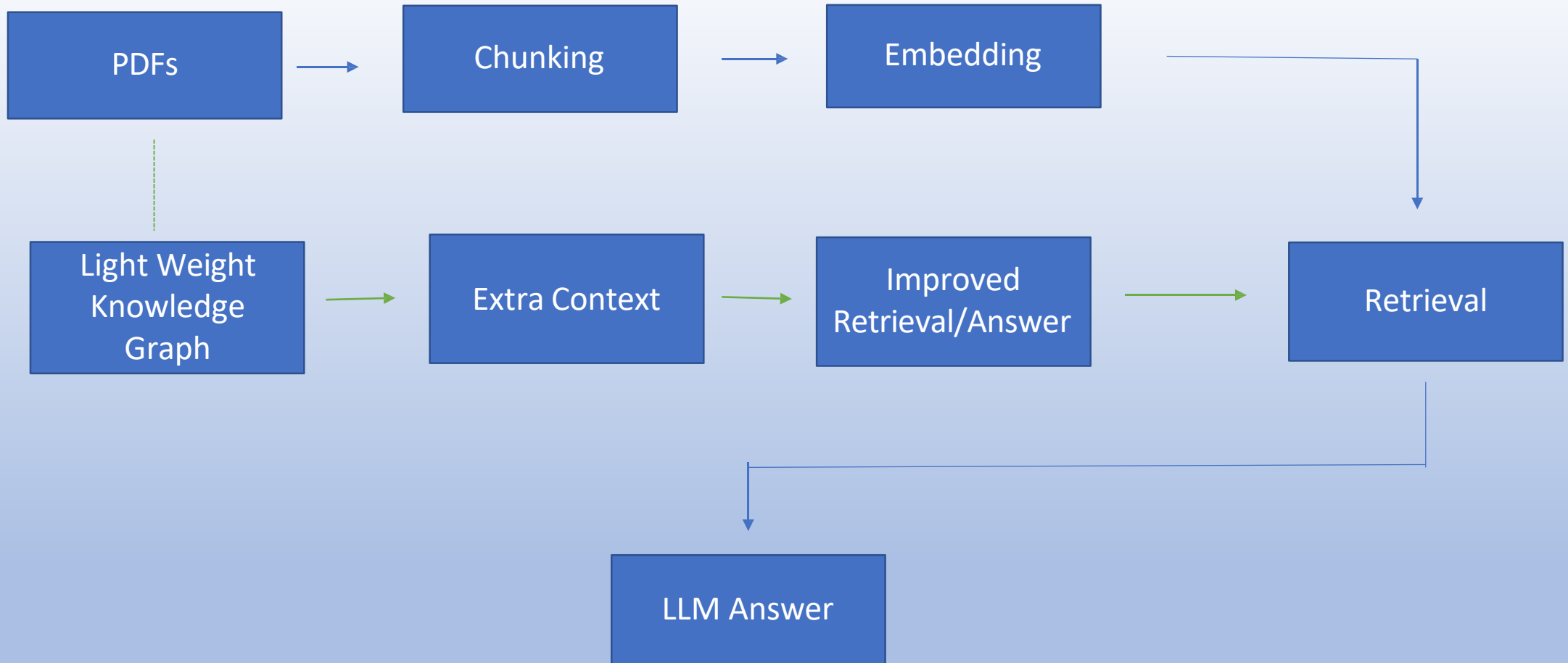
- The dataset consists of PDF lecture materials collected from the university portal. These documents are organized into three main course folders:
 - ❖ Machine Learning, Data Management, and Data Semantics.
- All PDFs are processed by extracting their text and splitting it into smaller chunks, which are then used as the retrieval corpus for the system.
- This makes the task highly domain-specific, ensuring that all answers are grounded in actual course materials rather than general open-domain knowledge.
- ❖ Overall, the dataset includes **118 PDF files** and **1048 extracted text chunks**.



Corpus Breakdown

- The dataset is distributed across three main courses, with the majority of content coming from Data Semantics.
- Machine Learning: 50 PDFs → 266 chunks
- Data Management: 17 PDFs → 83 chunks
- Data Semantics: 51 PDFs → 699 chunks
- This distribution highlights that Data Semantics contributes the largest portion of the corpus, which strongly influences retrieval and question answering behavior.

System Pipeline



Baseline RAG System

- The baseline system follows a standard Retrieval-Augmented Generation (RAG) pipeline.
- The user question is converted into an **embedding vector**.
- **ChromaDB** retrieves the most semantically similar text chunks from the dataset.
- The **LLM generates an answer using only the retrieved chunks**.
- This ensures responses are **grounded in course material** and easy to interpret.

Q. Why it is used as a baseline?

- Provides a **simple and transparent comparison point** for advanced methods.

❑ Key limitation

- Performance drops when:
 - Query wording differs from the PDFs
 - Answer requires combining multiple concepts or sections
- In curriculum QA, this is important because answers often depend on **cross-topic relationships**, not just single passages.



Knowledge Graph Enhancement

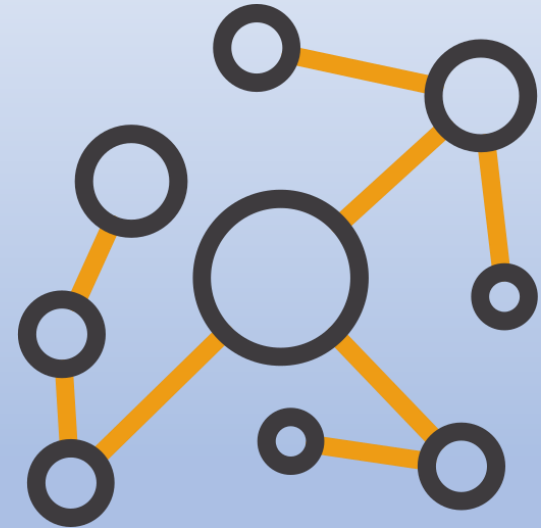
- The knowledge graph acts as a **lightweight semantic layer** over the RAG system.
- When a question matches a known topic, the system uses the graph to **add contextual information about related course concepts**.
- It links topics to their **course locations and semantically related ideas**, improving retrieval context.
- The graph does **not perform full symbolic reasoning**—its role is strictly supportive.

Q. How it works?

- Question is matched with known topics in the knowledge graph
- Graph expands context using:
 - Related course modules
 - Connected concepts (e.g., RDF → SPARQL, OWL)
- Retrieved text chunks are **enriched with graph-based context**
- Improves handling of **cross-topic curriculum questions**

Q. Why it helps?

- Improves retrieval when questions require **multiple related concepts**
- Reduces reliance on exact wording in PDFs
- Adds **structured curriculum awareness**



Examples

1. What is overfitting and how can we avoid it?

```
=====
QUESTION: What is overfitting and how can we avoid it?
=====

==> BASELINE RAG ANSWER:

According to the provided lecture content, overfitting is a situation where a classification model performs extremely well on the training set but poorly on new or unseen data (test set). This occurs when the model becomes too complex and starts to fit the noise in the training data rather than the underlying patterns.

To avoid overfitting, several techniques can be employed, including:

1. Regularization: This involves adding a penalty term to the loss function that encourages simpler models.
2. Data augmentation: This involves generating additional training data by applying random transformations or perturbations to the existing data.
3. Early stopping: This involves stopping the training process when the model's performance on the validation set starts to degrade.

These techniques can be found in the "Machine Learning" course, specifically in the file "Machine Learning - [2] Classification - 03 - Performance Evaluation - part I - 0 (1).pdf".

Additionally, feature engineering and handling outliers can also help prevent overfitting. Feature engineering is discussed in the "Data Management" course, specifically in the file "9. Data preparation.pdf".
```

2. What is overfitting and how can we avoid it?

```
=====
QUESTION: What is overfitting and how can we avoid it?
=====

==> KG-ENHANCED RAG ANSWER:

Based on the provided lecture content and knowledge graph context, overfitting is a type of problem that can occur in machine learning when a model becomes too specialized to the training data and does not generalize well to new, unseen data.

According to [Source 2] Course: Machine Learning | File: Machine Learning - [2] Classification - 03 - Performance Evaluation - part I - 0 (1).pdf, overfitting occurs when a model achieves a low training error but performs poorly on unseen data. This is illustrated by the blue line classification model, which has a low training error but incorrectly classifies many records in the test set.

To avoid overfitting, [Source 1] Course: Data Management | File: 9. Data preparation.pdf suggests several techniques, including:

* Imputation: replacing or filling in outlier values with estimated or imputed values
* Transformation: applying mathematical transformations to reduce the impact of outliers
* Winsorizing: capping or truncating extreme values at a certain threshold
* Data partitioning: partitioning the data into different subsets based on the presence of outliers

Additionally, [Source 3] Course: Machine Learning | File: Machine Learning - [2] Classification - 03 - Performance Evaluation - part I - 0 (1).pdf notes that overfitting can be avoided by using techniques such as regularization and early stopping.

Overall, overfitting is a common problem in machine learning that can lead to poor generalization performance. By understanding the causes of overfitting and using appropriate techniques to avoid it, we can build more accurate and reliable classification models.

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Knowledge Graph context used:
No KG context found for this question
-----

KG-enhanced retrieved 5 chunks
```

KG-enhanced RAG is valuable for relational questions, while baseline RAG remains strong for direct factual questions.

Related Work

This project is supported by three key foundational works:

1. Graph Retrieval-Augmented Generation: A Survey

- Main conceptual foundation for this work
- Defines GraphRAG as:
 - Graph-based indexing
 - Graph-guided retrieval
 - Graph-enhanced generation
- Directly informs the design of the KG-augmented retrieval layer

2. From Local to Global: A GraphRAG Approach to Query-Focused Summarization

- Demonstrates that graph-based indexing improves QA over large corpora
- Uses entity and graph structure to improve summarization quality
- Supports the idea that graph context improves retrieval beyond vector search

3. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

- Original RAG framework (baseline method)
- Introduces retrieve-then-generate architecture
- Serves as the benchmark baseline for comparison in this project



Evaluation Setup

❖ Evaluation Methodology

- 20 manually curated academic questions
- Questions cover:
 - Machine Learning
 - Data Management
 - Data Semantics
 - Cross-course topics

❖ Experimental Setup

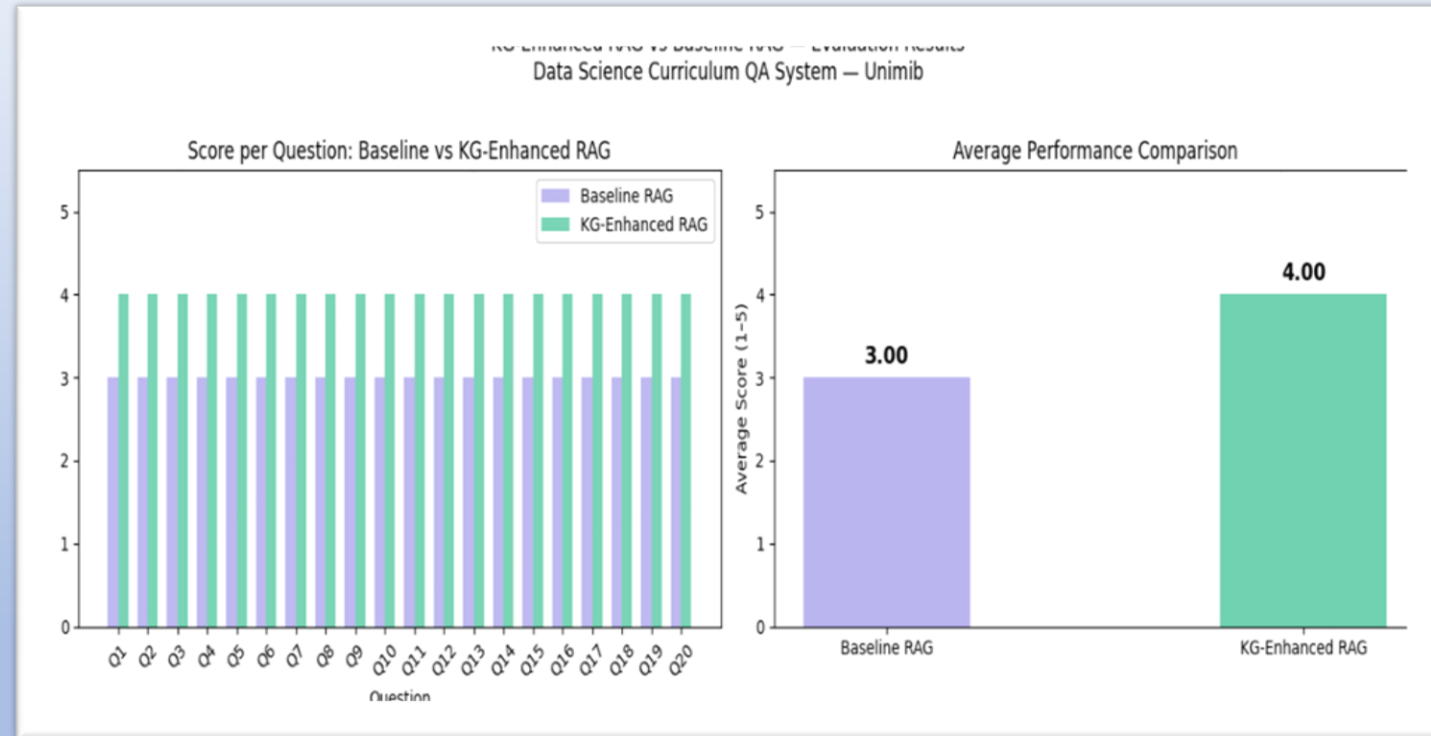
- Each question was answered twice:
 - **Baseline RAG**
 - **KG-Enhanced RAG**
 - **Evaluation Process**
- Answers stored in a CSV file
- Manual scoring used to compare answer quality
- Scores visualized using comparison charts



Evaluation Results

Evaluation Results:

- The **left chart** shows the average performance comparison between the two systems.
- The **baseline RAG** achieved an average score of **3.0**, while the **KG-enhanced RAG** achieved **4.0**.
- The **right chart** presents the question-by-question evaluation across all 20 questions.
- For every question, the **KG-enhanced RAG consistently scores higher than the baseline RAG**.



Evaluation Results

Metric	Baseline RAG	KG-Enhanced RAG
Average score	3.00 / 5	4.00 / 5
Relative improvement	—	+1.00 point
Strength	Direct semantic retrieval	Better curriculum context and topic linking
Weakness	Misses some relational clues	Depends on KG coverage

- KG-Enhanced RAG improves performance by **+1.00 point (3 → 4 / 5)**
- Gains come from **structured curriculum context + topic relationships**
- Trade-off: performance depends on **quality and coverage of KG**

Project Statistics

- Corpus & System Scale
- 118 PDF files processed
- 1,048 total text chunks extracted
 - 266 → Machine Learning
 - 83 → Data Management
 - 699 → Data Semantics
- 1,048 embeddings stored in ChromaDB
- 3 courses represented
- 3 topics indexed in Knowledge Graph



Discussion

❖ Strengths

- The project demonstrates how **knowledge graph structure improves retrieval quality** in a domain-specific academic QA system.
- The system is **practical and explainable**, making retrieval behavior easier to interpret.
- The graph is intentionally **lightweight rather than a full ontology**, keeping the implementation manageable for the project scope.
- The project scope was deliberately narrowed based on professor feedback to maintain depth and clarity.
- The primary focus is on **retrieval enhancement and answer quality**, not on complex RDF/SPARQL engineering.

❖ Limitations

- The knowledge graph covers only a limited set of topics.
- Questions outside graph coverage may not benefit from KG enhancement.
- Evaluation is manually scored and limited to 20 questions.

Future Improvement

❖ Possible Improvements

- Expand the knowledge graph with more topics and prerequisite links
- Include more diverse academic question types and a larger evaluation set
- Enhance retrieval using hybrid search (vector + lexical), query expansion, and graph-guided reranking

❖ Advanced Extensions

- Graph-based query expansion and reranking
- Topic-path reasoning across courses

❖ Project Direction

- Extensions should preserve focus and clarity
- Current system already meets course goals: focused, practical, explainable, and well-evaluated on academic QA tasks

Conclusion

- **KG-enhanced RAG improves answer quality over baseline RAG**
- **Uses real curriculum PDFs as the knowledge source**
- **Lightweight knowledge graph enhances retrieval effectiveness**
- **Evaluation shows consistent performance gains on curated academic questions**
- **The system is focused, practical, and explainable**
- **Designed in line with course guidance, avoiding unnecessary complexity**
- **Demonstrates the value of graph-assisted RAG in curriculum-specific QA**

Responsible AI usage

- **I used Perplexity for writing support, rewriting, coding assistance, and clarification.**
- **I used Perplexity to help refine the project explanation and presentation wording.**
- **I understand and can explain all major choices, code, and results**



CONCLUSION

Reference

- Peng, B., Zhu, Y., Liu, Y., Bo, X., Shi, H., Hong, C., Zhang, Y., & Tang, S. [Graph Retrieval-Augmented Generation: A Survey.](#)
- Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S., & Kiela, D. [Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.](#)
- Edge, D., Trinh, H., Cheng, N., Bradley, J., Chao, A., Mody, A., Truitt, S., Metropolitansky, D., Osazuwa Ness, R., & Larson, J. *From Local to Global: [A GraphRAG Approach to Query-Focused Summarization.](#)*